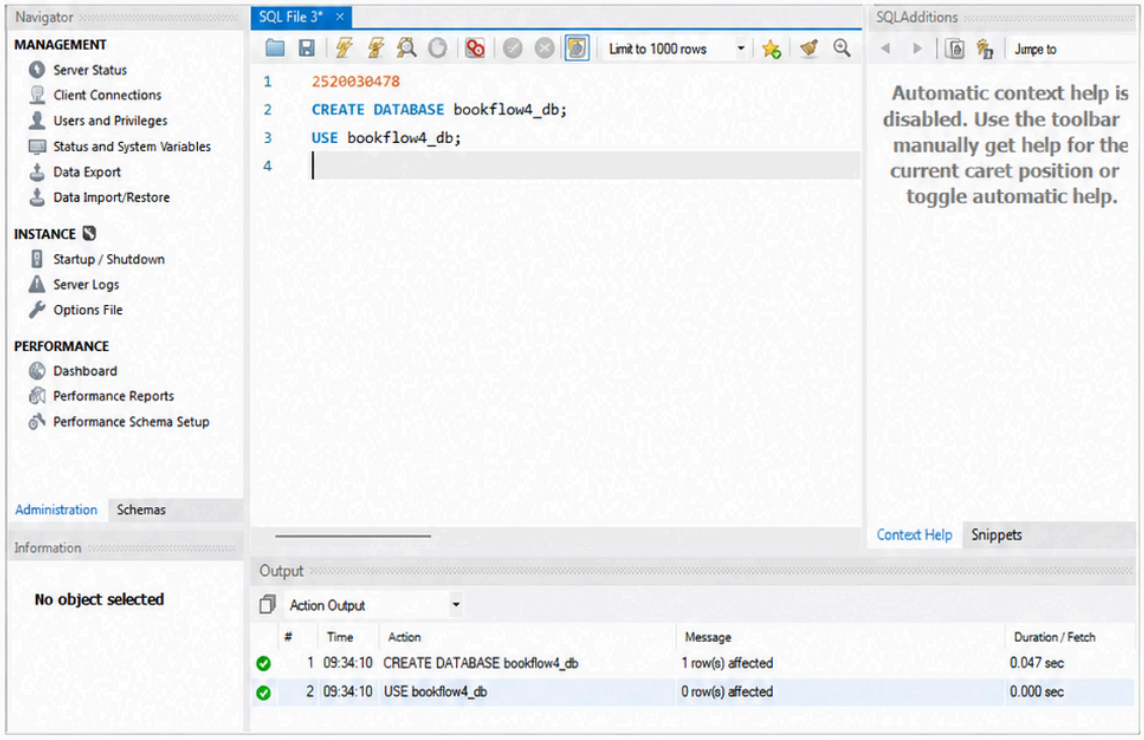
Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

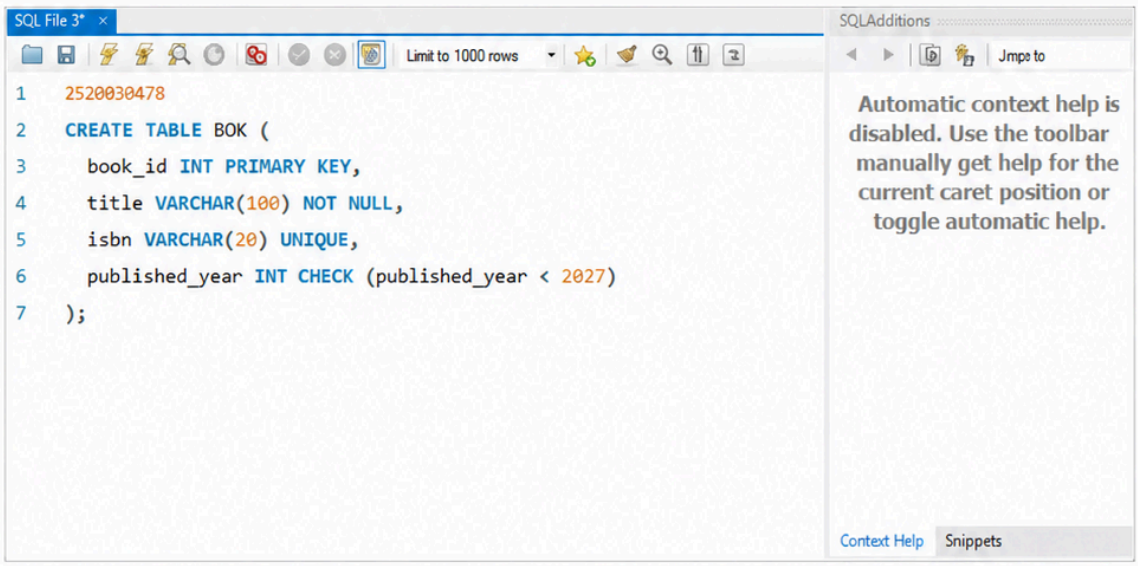
ROLL NO:2520030478

NAME: ADAMA SAMARTH

Query 1: -CREATE DATABASE



Query 2: Create Books Table (DDL + Constraints)



Output: -

✔	4	09:38:44	CREATE TABLE BOK (book_id INT PRIMARY KEY...	0 row(s) affected	0.109 sec

Query 3: Insert Books and display

```
11  2520030478
12  INSERT INTO Bok (book_id,title,isbn,published_year) VALUES
13  (1,'The Great Bay','9780743273565',2000),
14  (2,'The Indian Culture','2345987456',2010),
15  (3,'1947','9456783457934',2020);
16  SELECT * FROM Bok;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

book_id	title	isbn	published_year
1	The Great Bay	9780743273565	2000
2	The Indian Culture	2345987456	2010
3	1947	9456783457934	2020
NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types

Output:-

✔	12	12:10:49	INSERT INTO Bok (book_id,title,isbn,published_y...	3 row(s) affected Records: 3 Duplicates: 0 Warnings:	0.062 sec
✔	13	12:10:49	SELECT * FROM Bok LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Query 4: Members Table: Create

```
1  CREATE TABLE Members(
2   member_id INT PRIMARY KEY,
3   full_name VARCHAR(100),
4   email VARCHAR(100) UNIQUE
5  );
```

Output:-

✔	19	12:20:46	CREATE TABLE Members(member_id INT PRIM...	0 row(s) affected	0.063 sec

Query 5:- Insert Members table

```sql
INSERT INTO Members (member_id, full_name, email) VALUES
(101, 'Adama Samarth', 'adamasamarth@gmail.com'),
(102, 'Rohit Verma', 'rohitverma@gmail.com'),
(103, 'Vivek Sharma', 'viveksharma@gmail.com');
```

Output: -

✔	19	12:20:46	INSERT INTO Members (member_id,full_name,email) VALUES ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

Query 6: - : Display Members table

```sql
SELECT * FROM Members;
```

Output: -

Result Grid | Filter Rows: | Edit: | Export/Import:

member_id	full_name	email
101	Adama Samarth	adamasamarth@gmail.com
102	Rohit Verma	rohitverma@gmail.com
103	Vivek Sharma	viveksharma@gmail.com
NULL	NULL	NULL

Result Grid | Form Editor | Field Types

Members 3 | Apply | Revert | Context Help | Snippets

Output: -

✔	20	12:22:31	INSERT INTO Members (member_id,full_name,email) VALUES ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
✔	21	12:24:30	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Query 7: Create Loans Table

```sql
CREATE TABLE Loans (
  loan_id INT PRIMARY KEY,
  member_id INT NOT NULL,
  amount DECIMAL(10,2) NOT NULL CHECK (amount > 0),
  loan_date DATE NOT NULL,
  FOREIGN KEY (member_id) REFERENCES Members(member_id)
);
```

Output: -

✔	22	12:26:05	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_...	0 row(s) affected	0.078 sec

2520030478-Adama Samarth

```
CREATE TABLE Loan (
  load_id INT PRIMARY KEY,
  member_id INT,
  book_id INT,
  loan_date DATE,
  FOREIGN KEY (member_id) REFERENCES Members(member_id),
  FOREIGN KEY(member_id) REFERENCES Bok(book_id)
);
```

Output:-

```
23 12:33:08 CREATE TABLE Loan (load_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIG... 0 row(s) affected
```

Query 8: Insert 10 Loans

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

Output: -

```
91 13:41:27 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, ... 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 0.04 sec
```

Query 9: Display

```
SELECT * FROM Loans;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import:

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	103	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01
NULL	NULL	NULL	NULL

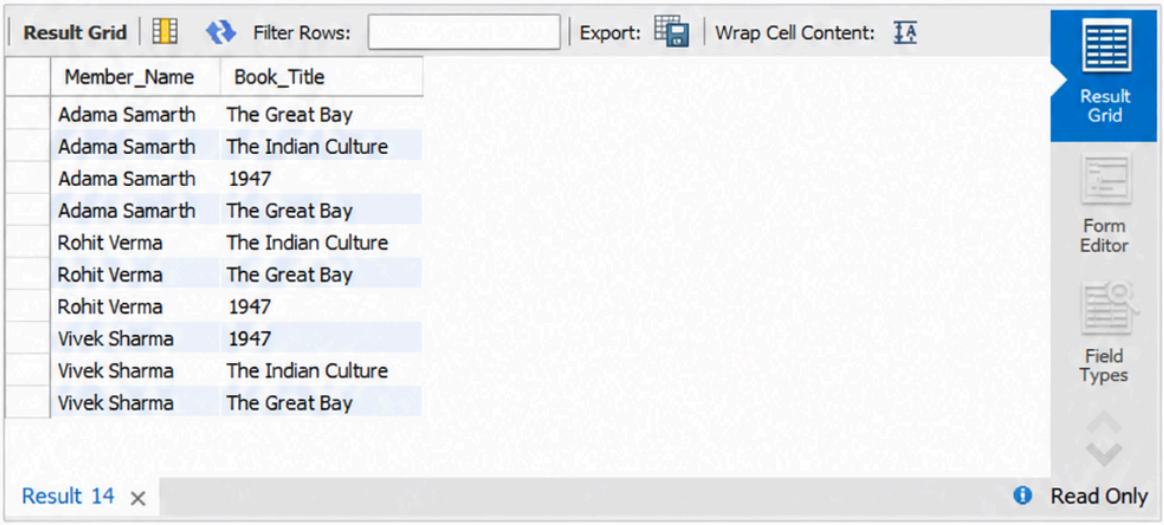
Result Grid | Form Editor | Field Types

Loans 7 | Apply | Revert

Query 10: JOIN Query

```
45    SELECT m.full_name AS Member_Name, b.title AS Book_Title
46    FROM loans l
47    INNER JOIN Members m ON l.member_id = m.member_id
48    INNER JOIN Bok b ON l.book_id = b.book_id;
```

Output: -



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Member_Name	Book_Title
Adama Samarth	The Great Bay
Adama Samarth	The Indian Culture
Adama Samarth	1947
Adama Samarth	The Great Bay
Rohit Verma	The Indian Culture
Rohit Verma	The Great Bay
Rohit Verma	1947
Vivek Sharma	1947
Vivek Sharma	The Indian Culture
Vivek Sharma	The Great Bay

Result Grid | Form Editor | Field Types | Read Only

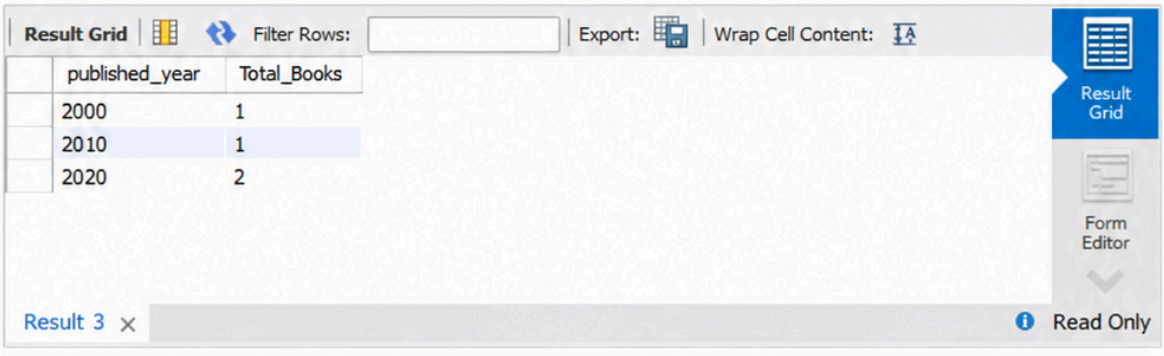
Result 14

✓	103	13:50:44	SELECT m.full_name AS Member_Name,b.title AS Book_Title FROM loans l INNER JOIN Members m ON l.member_id = m.member_id ∄ INNER JOIN Bok b ON l.book_id = b .book_id	10 row(s) returned

Query 11: GROUP BY Query

```
1    SELECT published_year, COUNT(book_id) AS Total_Books
2    FROM Bok
3    GROUP BY published_year
4    ORDER BY published_year;
```

Output: -



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

published_year	Total_Books
2000	1
2010	1
2020	2

Result Grid | Form Editor | Read Only

Result 3

✓	104	13:51:12	SELECT published_year, COUNT(book_id) AS Total_Books FROM Bok GROUP BY published_year ORDER BY published_year	3 row(s) returned